Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form		
Lab Name	Iterative Statements (Loops) in C Programming	
Lab Number	06	
Submitted by:		
Student Names		**Registration #**
Sumaira Safeer		FA22-BCE-019

Rubrics		**Marks**		
		1	**2**	**3**
PLO-5 **Modern Tools Usage**	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 **Communication Skills**	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab 06: Iterative Statements (Loops) in C Programming

Objective

To understand and implement iterative statements (while and for loops) in C programming for solving repetitive computational problems such as series generation, factorial calculation, number conversion, and mathematical operations.

Software Requirements

- Code::Blocks IDE / Dev-C++
- GCC Compiler.
  - Windows Operating System.

Experiments Performed

In-Lab Tasks

1. Print even numbers from 2 to 100 using a while loop.
2. Calculate the power of a number using repeated multiplication.
3. Display Armstrong numbers between 1 and 500.

Post-Lab Tasks

1. Calculate the factorial of a number using a while loop.
2. Convert a decimal number into its octal equivalent.
3. Calculate overtime pay for ten employees based on overtime hours worked.

Learning Outcomes

After completing this lab, the student will be able to:

- Implement while and for loops in C.
- Solve repetitive mathematical problems using iteration.
- Generate numerical sequences.
- Perform number system conversions.
- Calculate factorials and powers using loops.
- Develop practical applications involving iterative processing.

In-Lab Tasks:

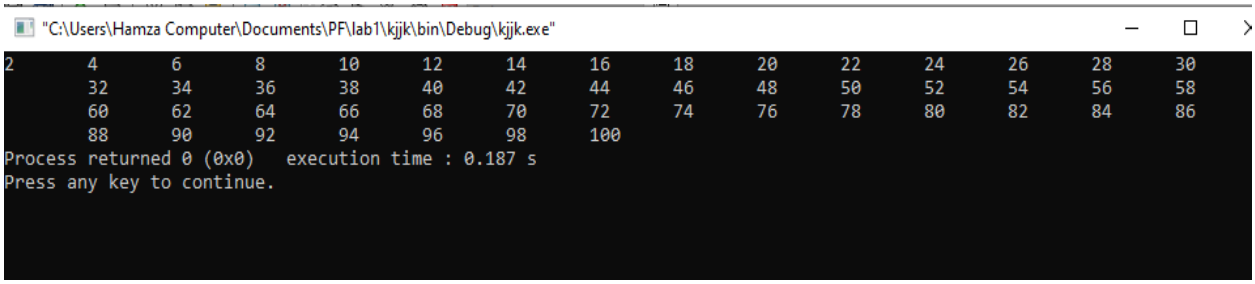
Task1:

Print even numbers from 2 to 100 using a while loop.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
   int number=2;
   while (number<=100)
      {
      printf("%d\t",number);
      number+=2;
      }
   return 0;
}
```

Output:



Task2:

Calculate the power of a number using repeated multiplication.

Code:

```
#include <stdio.h>
#include <stdlib.h>
```

```
int main()
{
   int Base, Exponent;
   int result = 1;
   printf("Enter the Base number: ");
   scanf("%d", &Base);

   printf("Enter the Exponent: ");
   scanf("%d", &Exponent);

   while (Exponent > 0) {
      result *= Base;
      Exponent--;
   }
   printf("Result: %d\n", result);
   return 0;
 }
```

Output:

```
Enter the Base number: 564
Enter the Exponent: 3
Result: 179406144

Process returned 0 (0x0)   execution time : 6.745 s
Press any key to continue.
```

Task3:

Display Armstrong numbers between 1 and 500.

Code:

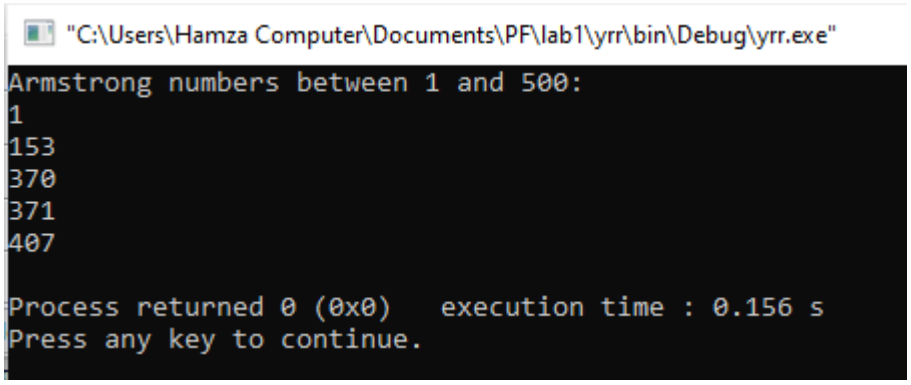
```
#include <stdio.h>
#include <stdlib.h>
int main()
{
```

```
int Number, original_Num, remainder, result;
   printf(“Armstrong numbers between 1 and 500:\n”);

   for (Number=1; Number<=500; Number++) {
      original_Num= Number;
      result = 0;
      while (original_Num!=0) {

         remainder = original_Num%10;
         result += pow(remainder, 3);
         original_Num /= 10;
      }
      if (result == Number) {
         printf(“%d\n”, Number);
      } }
    return 0;
   }
```

Output:



Post-lab Tasks:

Task1:

Calculate the factorial of a number using a while loop.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
   int number;
   int factorial = 1;
   printf("Enter a number: ");
   scanf("%d", &number);
   int i = 1;
   while (i <= number) {
      factorial *= i;
      i++;
   }
   printf("Factorial of %d is %d\n", number, factorial);
   return 0;
  }
```

Task2:

Convert a decimal number into its octal equivalent.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
   int number, original_Num, remainder;
```

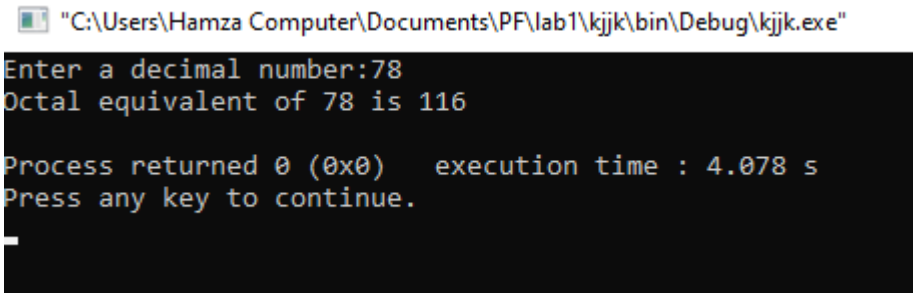
```
    int octal= 0, Value= 1;

    printf("Enter a decimal number:");
    scanf("%d", &number);

    original_Num = number;
    while (number != 0) {
        remainder = number%8;
        octal += remainder * Value;
        number/= 8;
        Value *= 10;
    }

    printf("Octal equivalent of %d is %d\n", original_Num, octal);
    return 0;
}
```

Output:



Task3:

Calculate overtime pay for ten employees based on overtime hours worked.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
```

```
int Employee= 1;
int hours_Worked, Overtime_Hours;
int Overtime_Rate=12, Regular_Hour=40;
 float Overtime_Pay;

while (Employee <= 10) {
    printf("Enter number of hours worked for Employee %d=", Employee);
    scanf("%d", &hours_Worked);

    if (hours_Worked > Regular_Hour) {
        int overtime_Hours = hours_Worked - Regular_Hour;
        Overtime_Pay = Overtime_Hours * Overtime_Rate;
    } else {
        Overtime_Pay = 0;
    }
    printf("Overtime Pay for Employee %d= Rs.%f \n", Employee, Overtime_Pay);
    Employee++;
}
return 0;
```